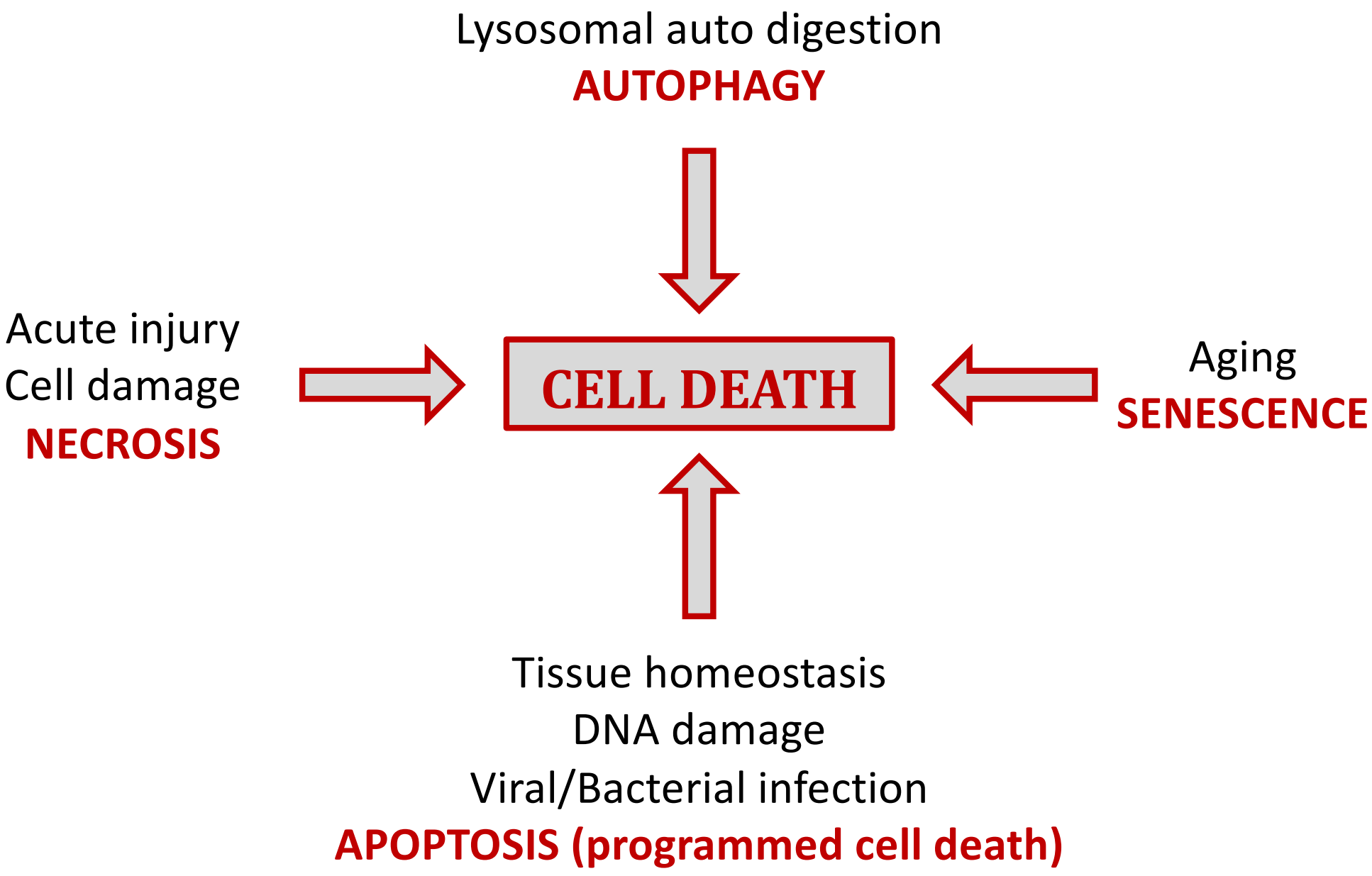
Apoptosis and senescence

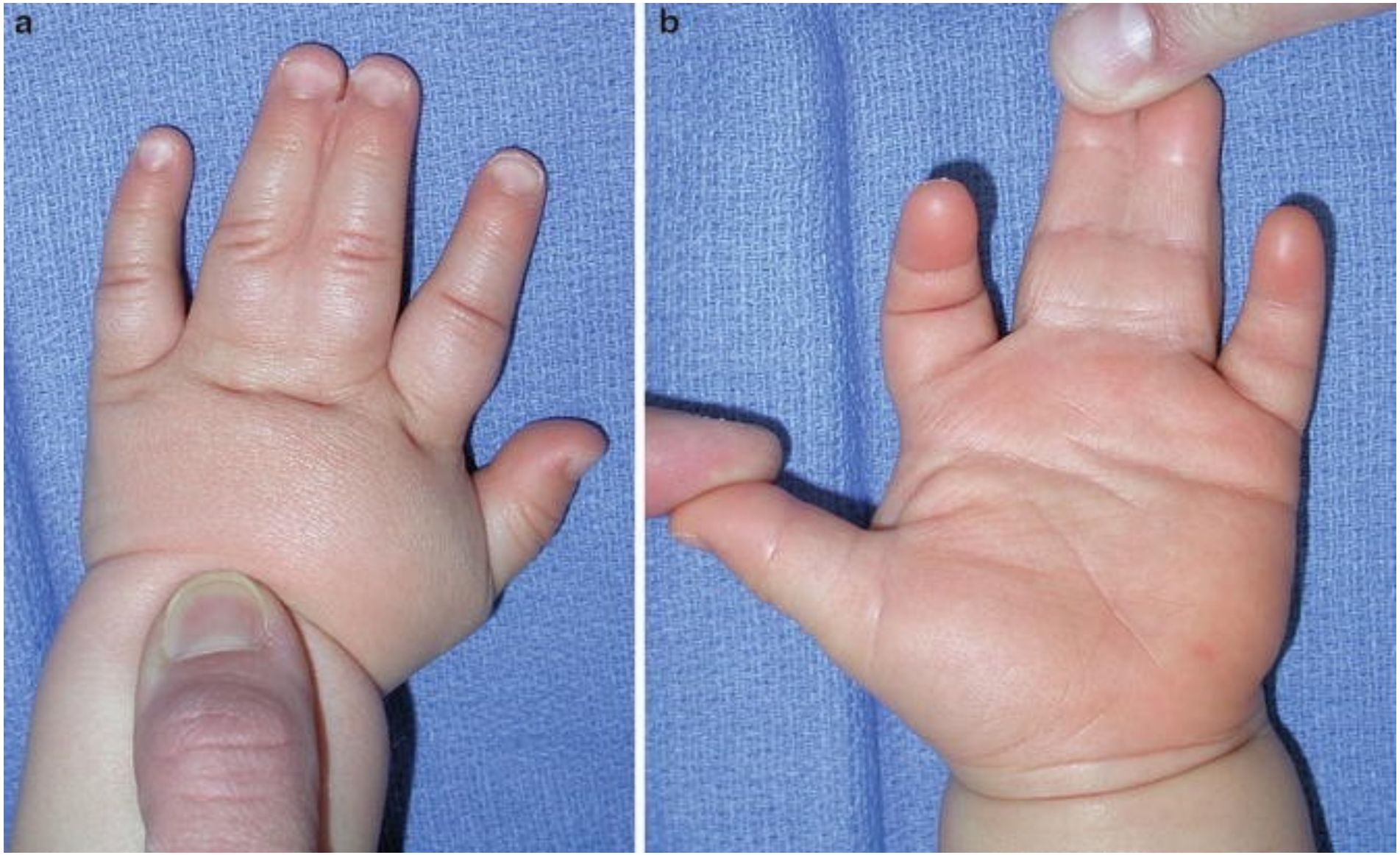
TBL 7.1

OCTOBER 16th, 2026

Dr. Serrano | dserrano@unav.es


Lysosomal auto digestion
AUTOPHAGY
Acute injury
Cell damage
NECROSIS
CELL DEATH
Aging
SENESCENCE
Tissue homeostasis
DNA damage
Viral/Bacterial infection
APOPTOSIS (programmed cell death)

APOPTOSIS



APOPTOSIS

-Controlled cell death: 60 billion cells per day. Elimination of unnecessary cells.

-Balance proliferation-apoptosis in healthy tissues

-Cancer: ↑ Mitotic / apoptotic ratio

-Ensures genome integrity

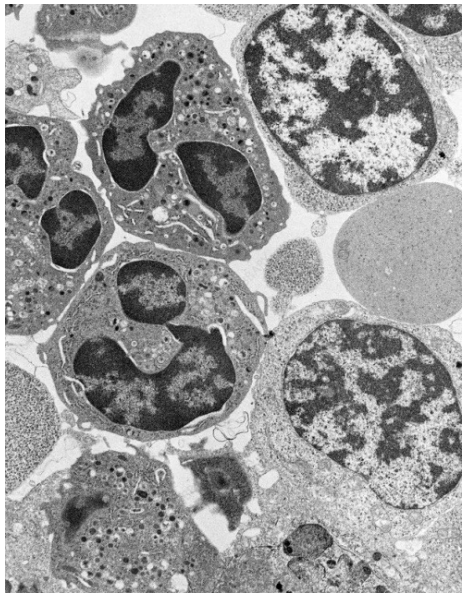
-Genetically controlled => cell suicide for the common good…

-Apoptosis involves:

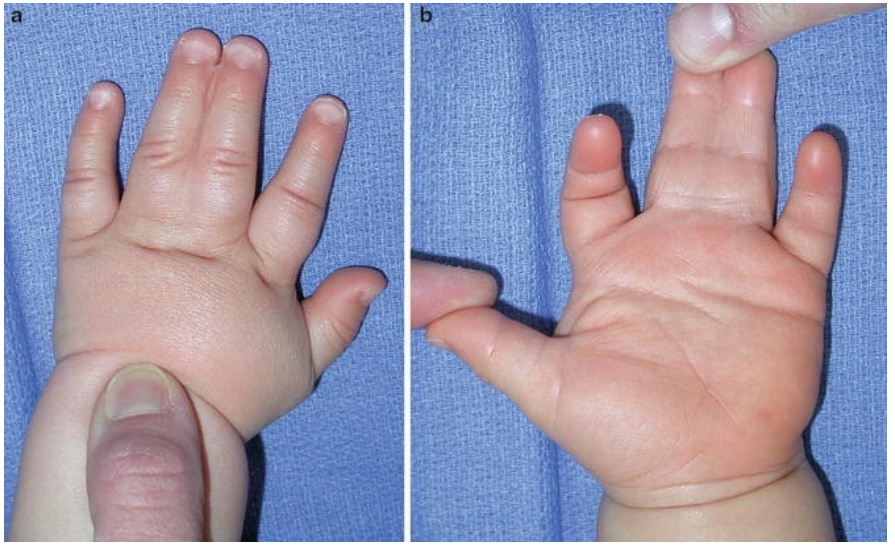
-Genetic / biochemical changes

-Morphological changes

LIVING CELLS **APOPTOSIS**







Syndactyly

Apoptosis in light microscopy images

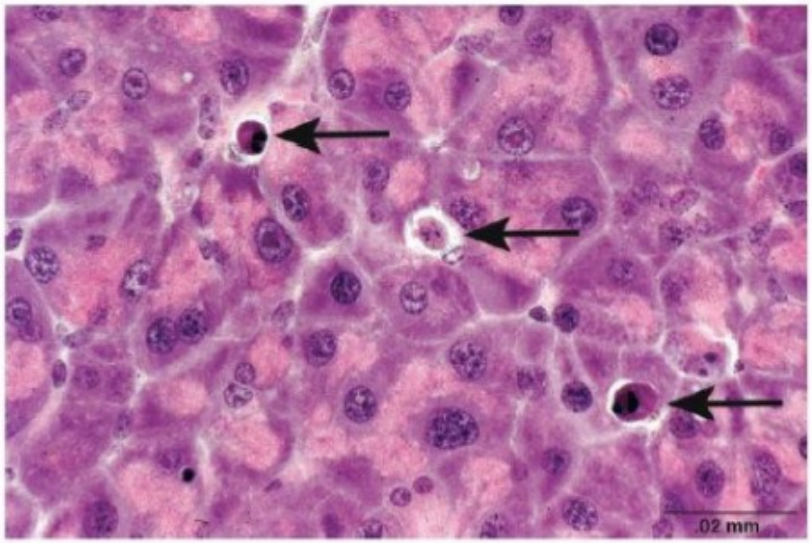
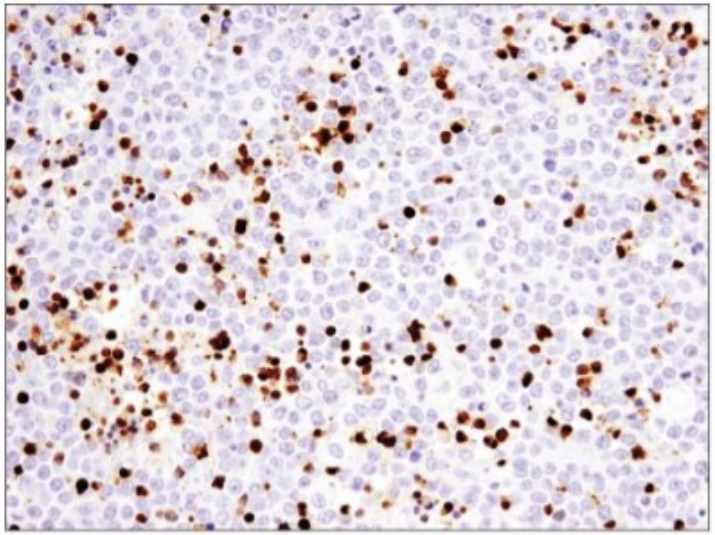
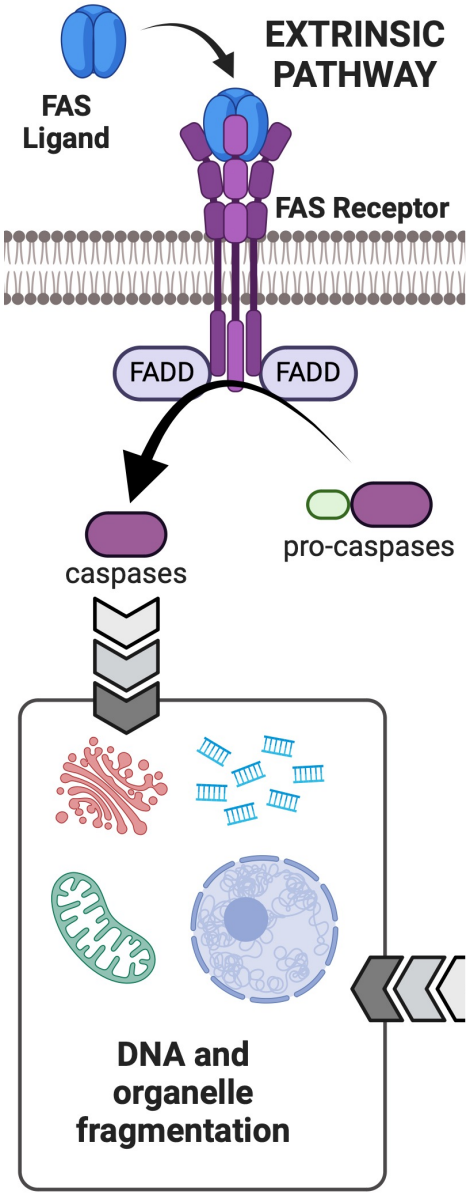


Figure 2. Apoptosis of exocrine pancreatic cells with cytoplasmic and nuclear condensation and nuclear fragmentation (arrows). Image courtesy of National Toxicology Program (NTP) archives.

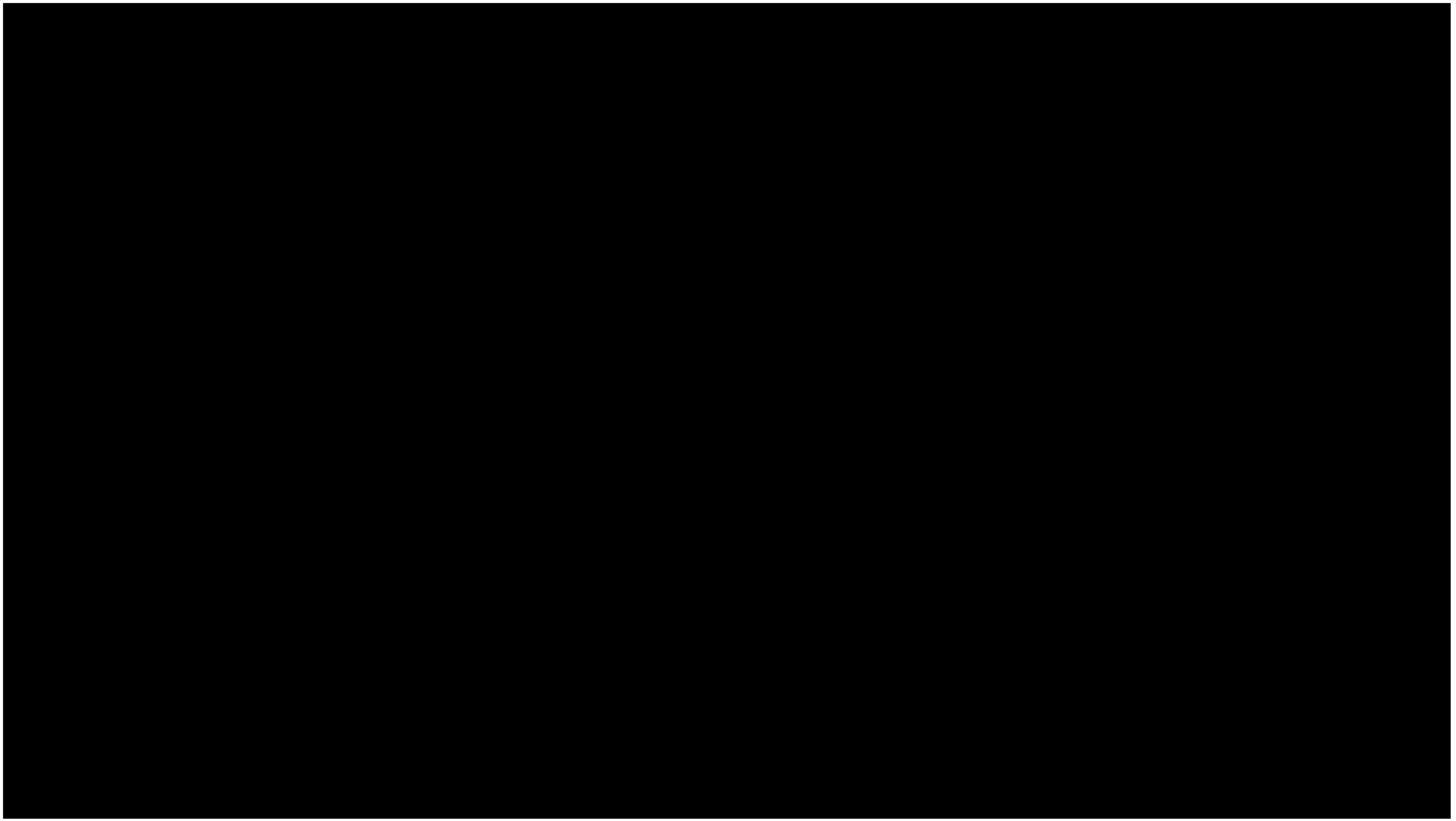


Apoptotic cells (brown) identified by Immunohistochemistry of apoptotic markers, in a section of Hodgkin lymphoma

APOPTOSIS ACTIVATION - OVERVIEW

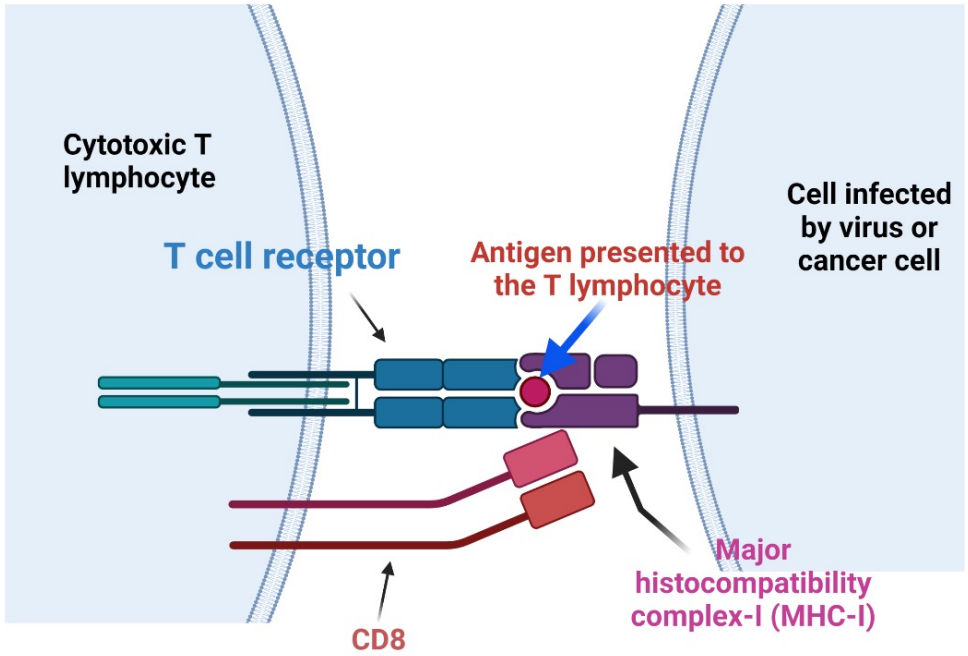


APOPTOSIS ACTIVATION

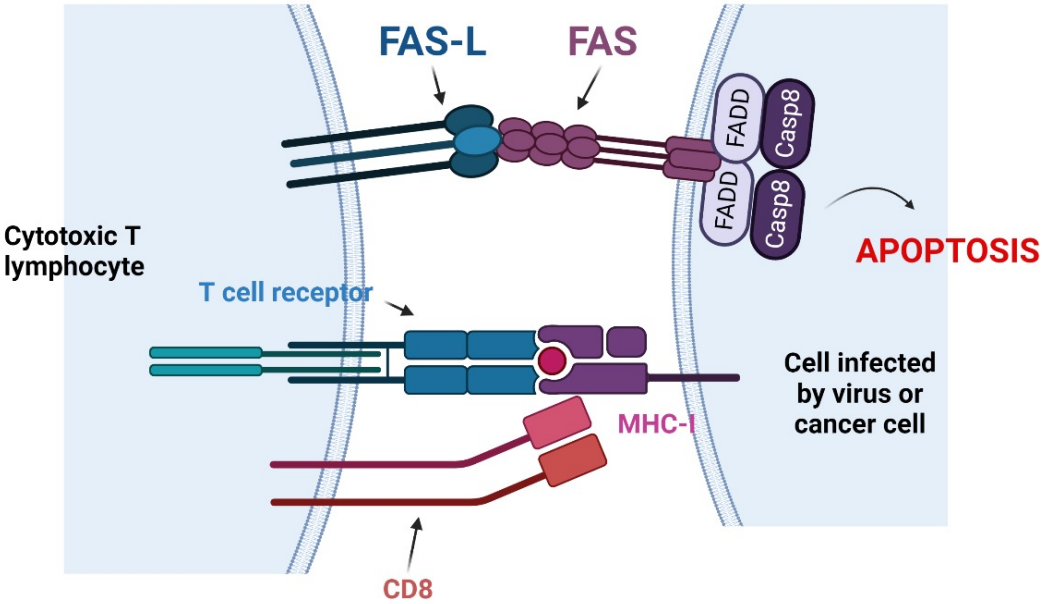


https://www.youtube.com/watch?v=-vmtK-bAC5E

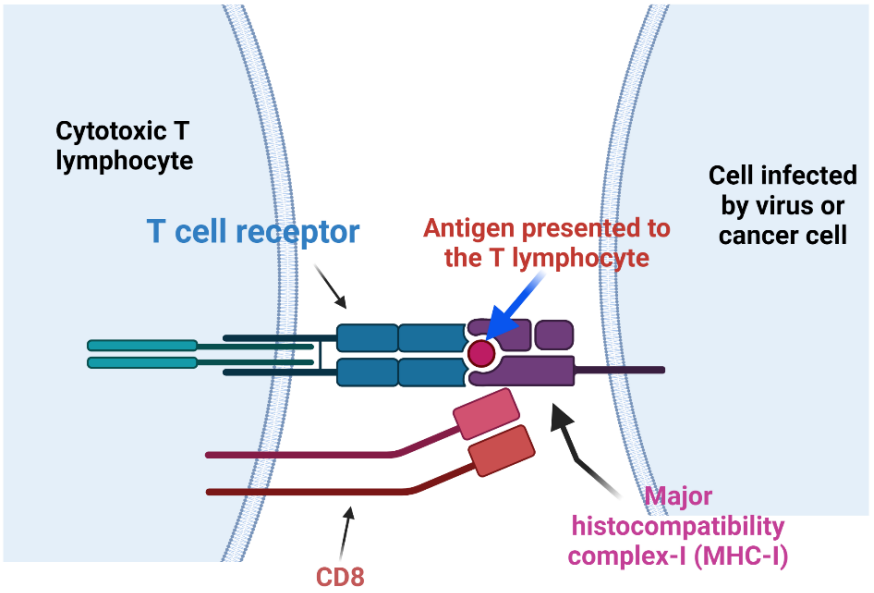
How T lymphocytes kill cells by inducing apoptosis



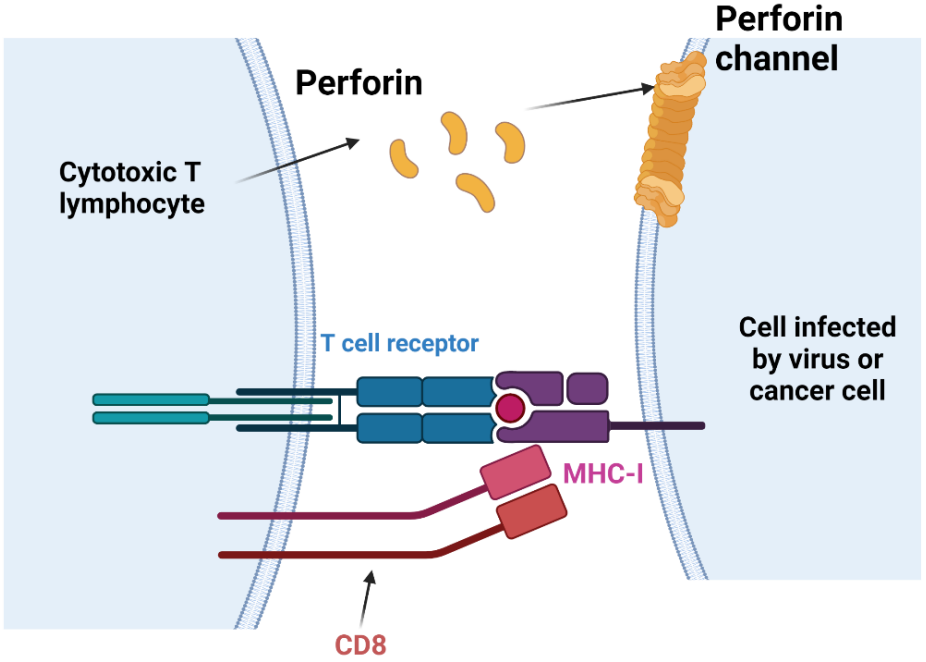
a) Extrinsic pathway

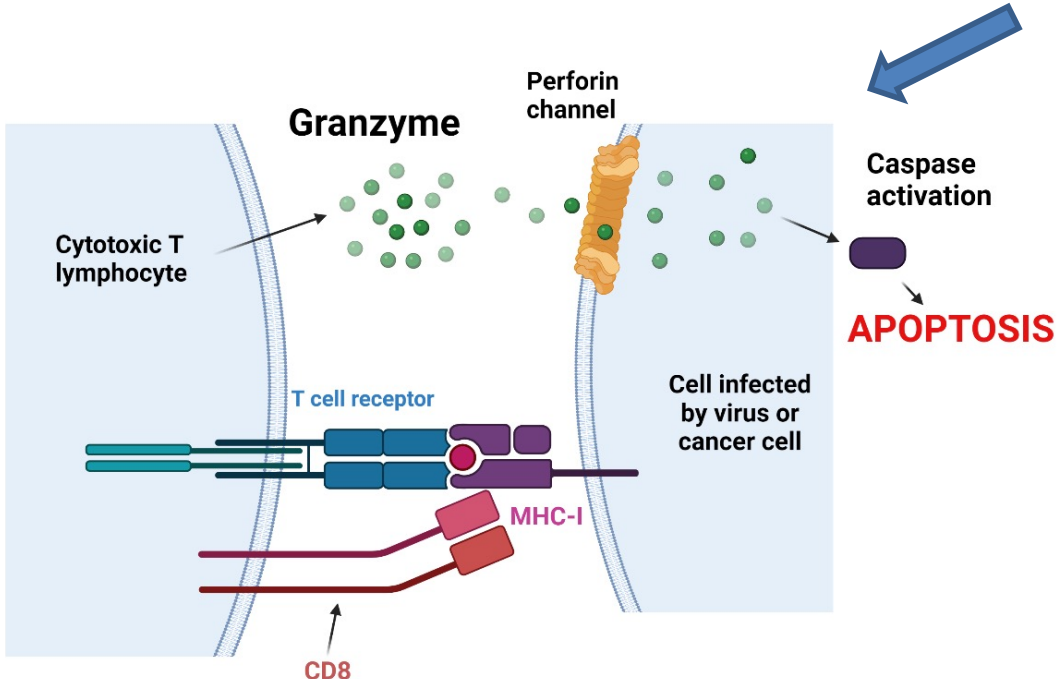


How T lymphocytes kill cells by inducing apoptosis



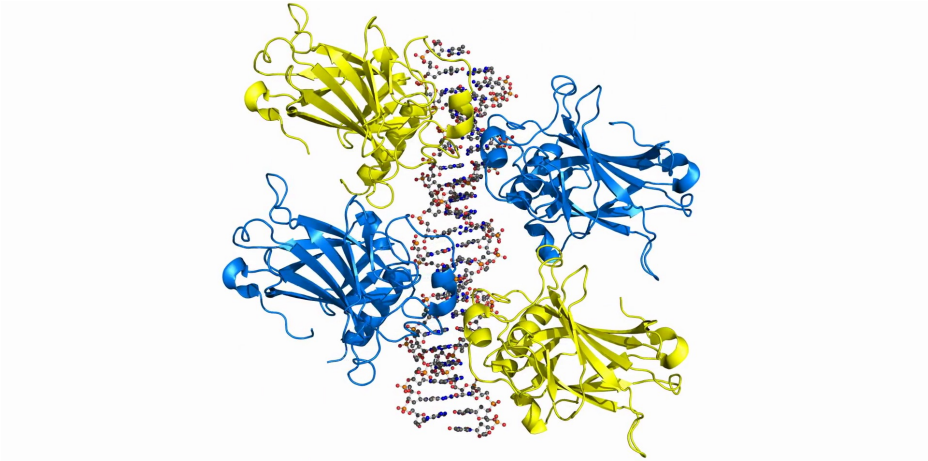
b) Intrinsic pathway





https://www.youtube.com/watch?v=WdCiaIS2LV4

p53: THE GUARDIAN OF THE GENOME



-53kDa protein

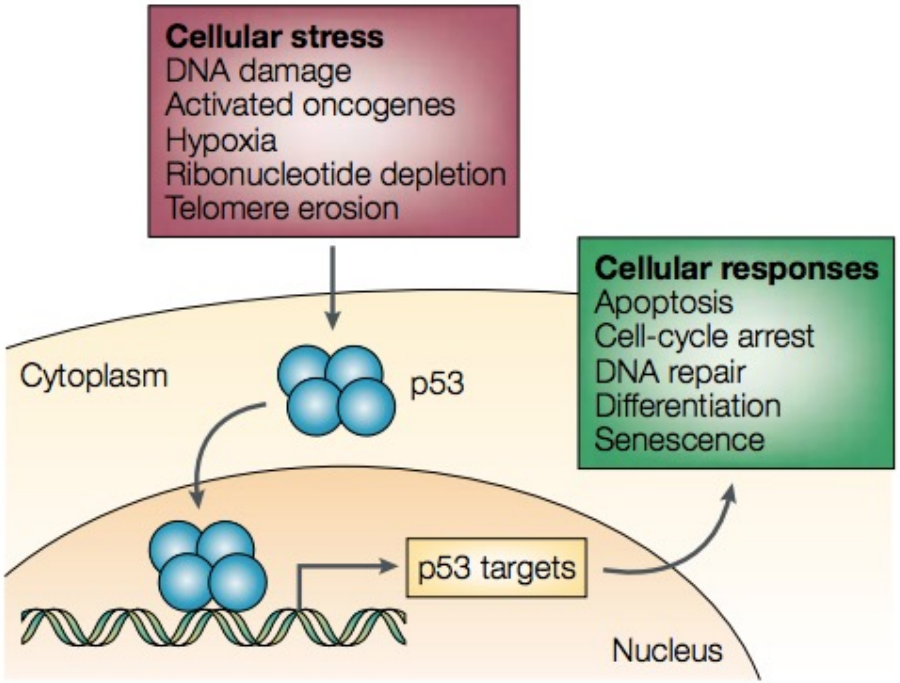
-Transcription factor

-Activated by phosphorylation

-Main regulator: Mdm2

-Homotetramer when active

50% of tumors harbor p53 mutations!



<u>Inactivated by Mdm2:</u>

-Inhibits p53 gene regulation activity

-Exports p53 out of the nucleus

-Induces p53 degradation (by ubiquitination)

Up-regulated in most cancers!

SENESCENCE

Telomere shortening

GENETIC FACTORS

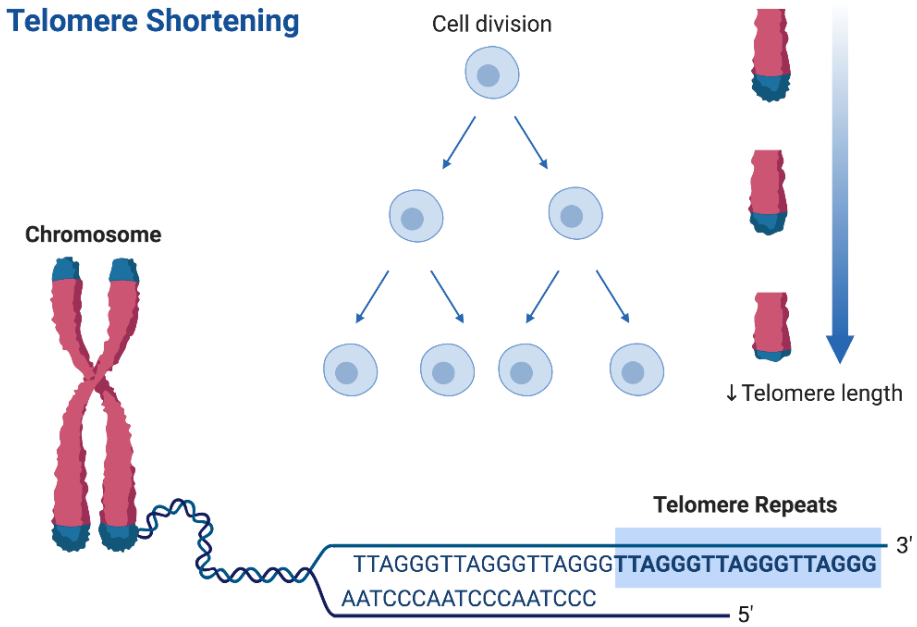
SENESCENCE

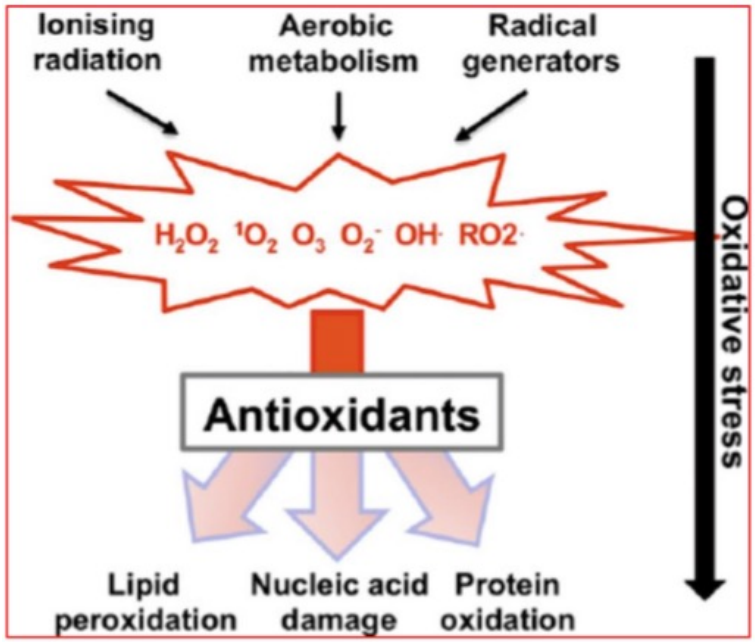
Loss of proliferation capacity

Permanently arrested in G1

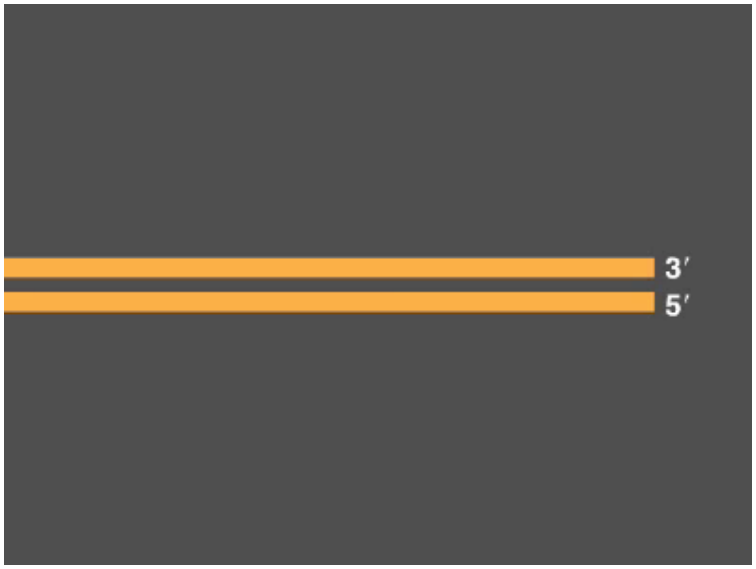
OXIDATIVE STRESS AND EXTERNAL DAMAGE

Alteration of cell molecules

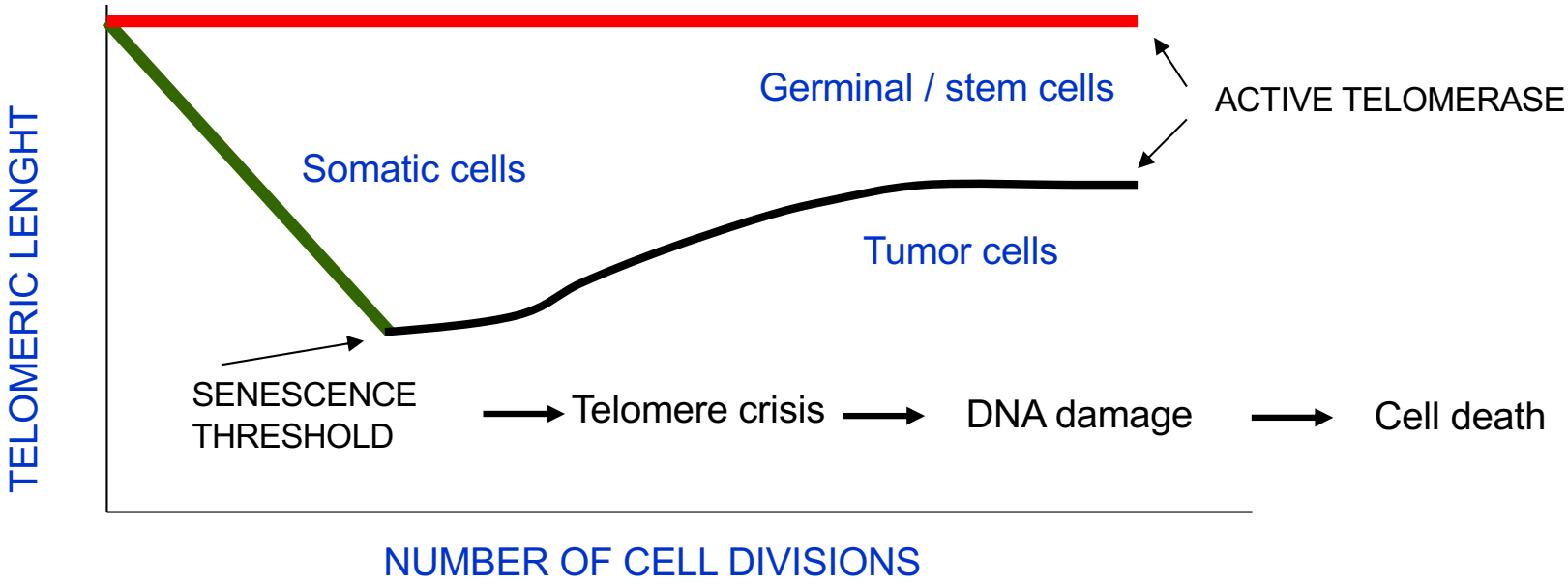




TELOMERE SHORTENING



https://www.youtube.com/watch?v=2NS0jBPurWQ



PROGERIA – Hutchinson-Gilford Syndrome

-Single mutation in **Lamin-A:** Premature aging (progeria)

-Low incidence: ~200 cases in the world

-Life expectancy: 13 years

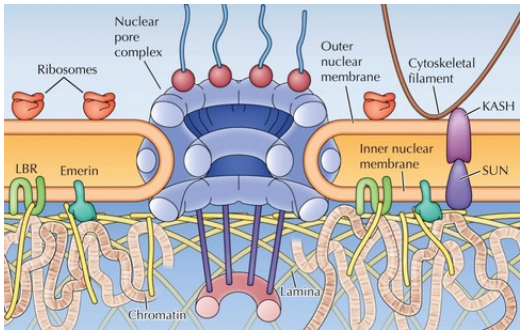
-At age 13, the body resembles a person > 100 years old



Mutations in **Lamin A** (LMNA)
(Structural protein in the nucleus)

??

Accelerated aging

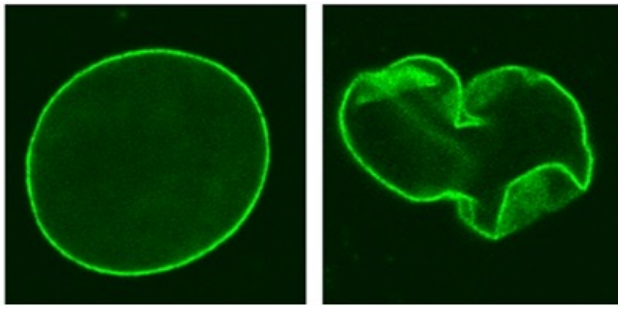


-Impairment of nuclear transport

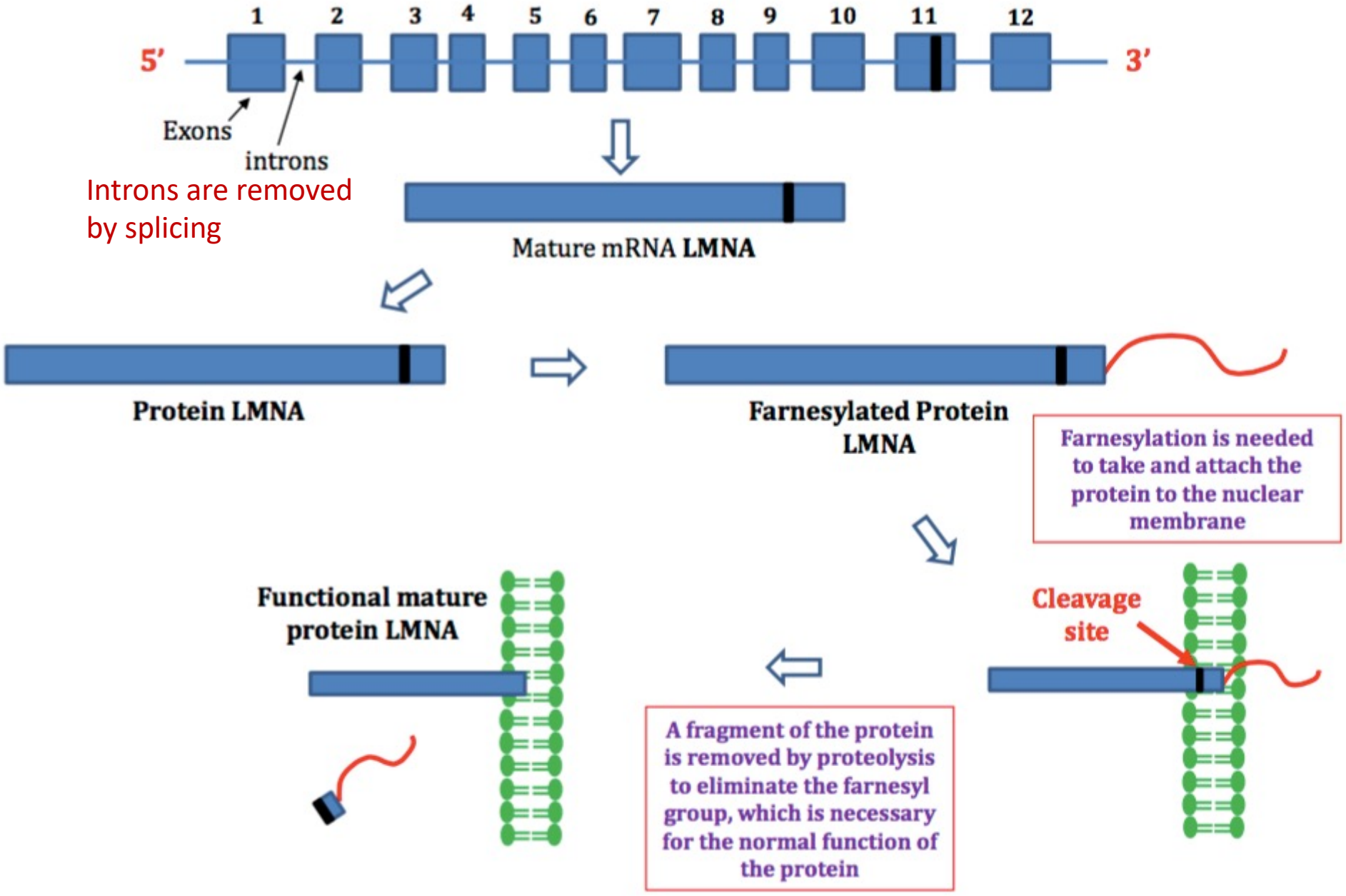
-Alteration of gene function

-Telomere shortening

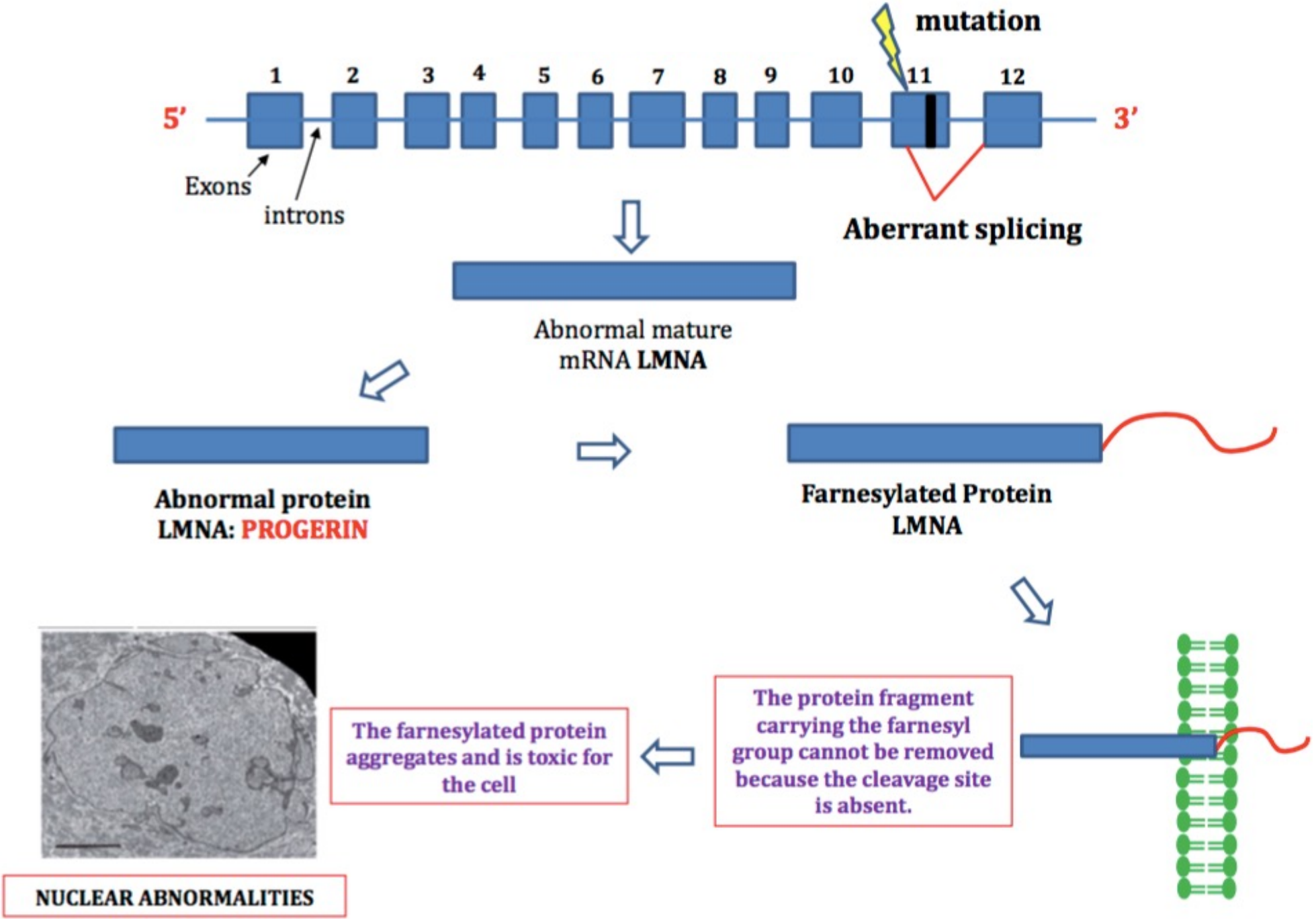
Normal Progeria



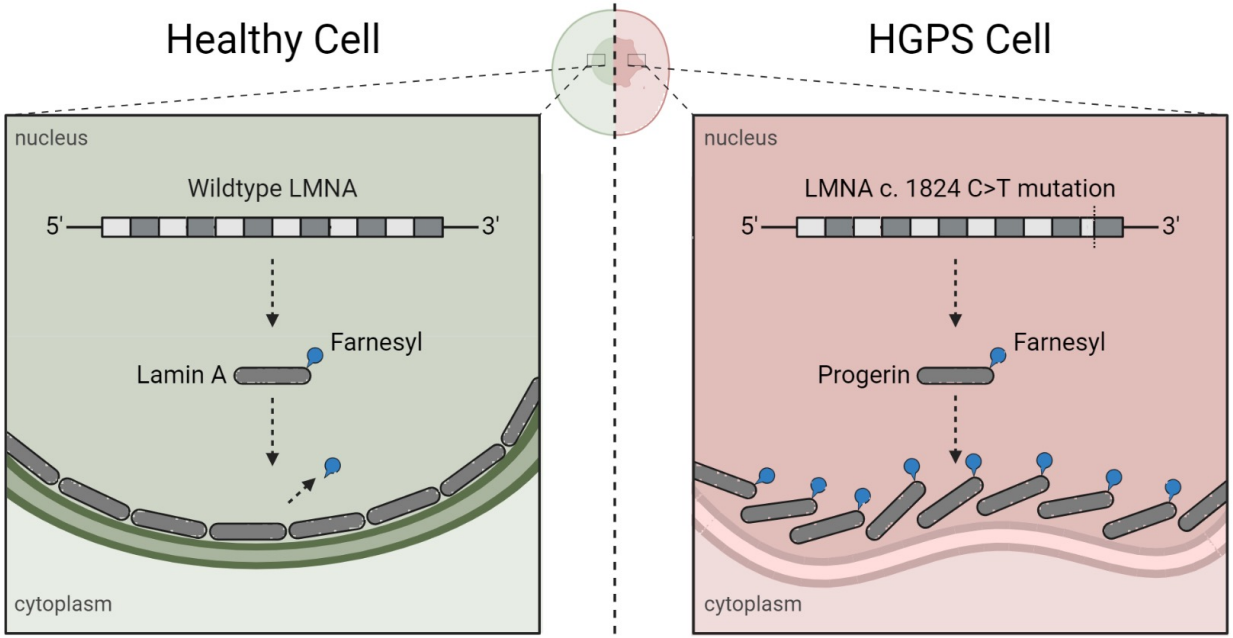
Normal LMNA gene



Mutated LMNA gene



The presence of progerin creates nuclear stress and accelerated aging (not well-defined mechanisms)

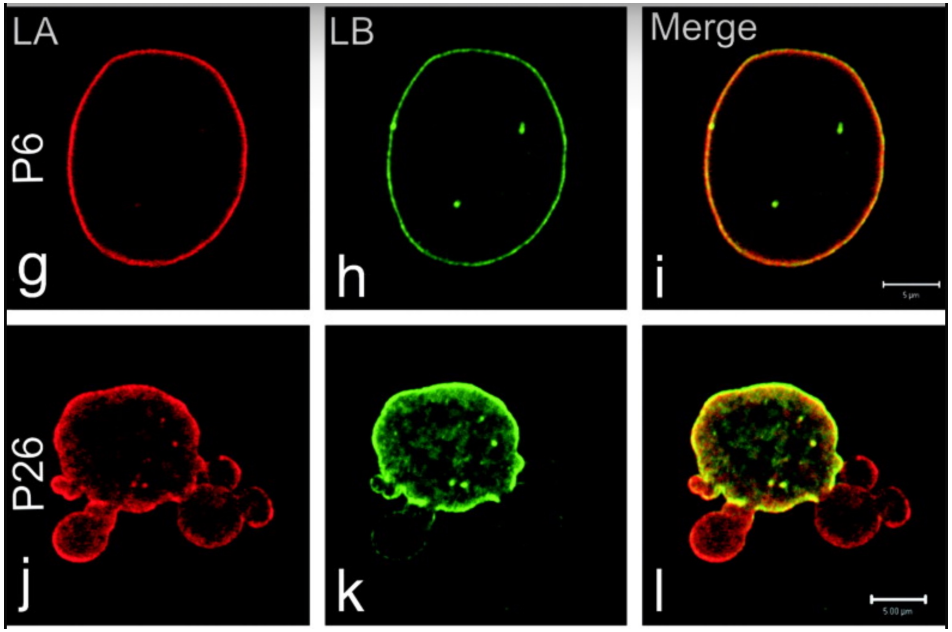


DNA damage

Defective DNA repair

-p53 activation
-cellular and tissular senescence

Accelerated aging



Lamin B
Lamin A: Mutant

CUTTING EDGE THERAPIES

Lonafarnib: Farnesyltransferase inhibitor

The protein does not get farnesylated and toxicity is reduced, but LMNA does not get into the membrane correctly...

"Gene corrector" (CRISPR-Cas9) and gene therapy

Experiments in transgenic mice where progeria was generated improved when this system was delivered in different organs



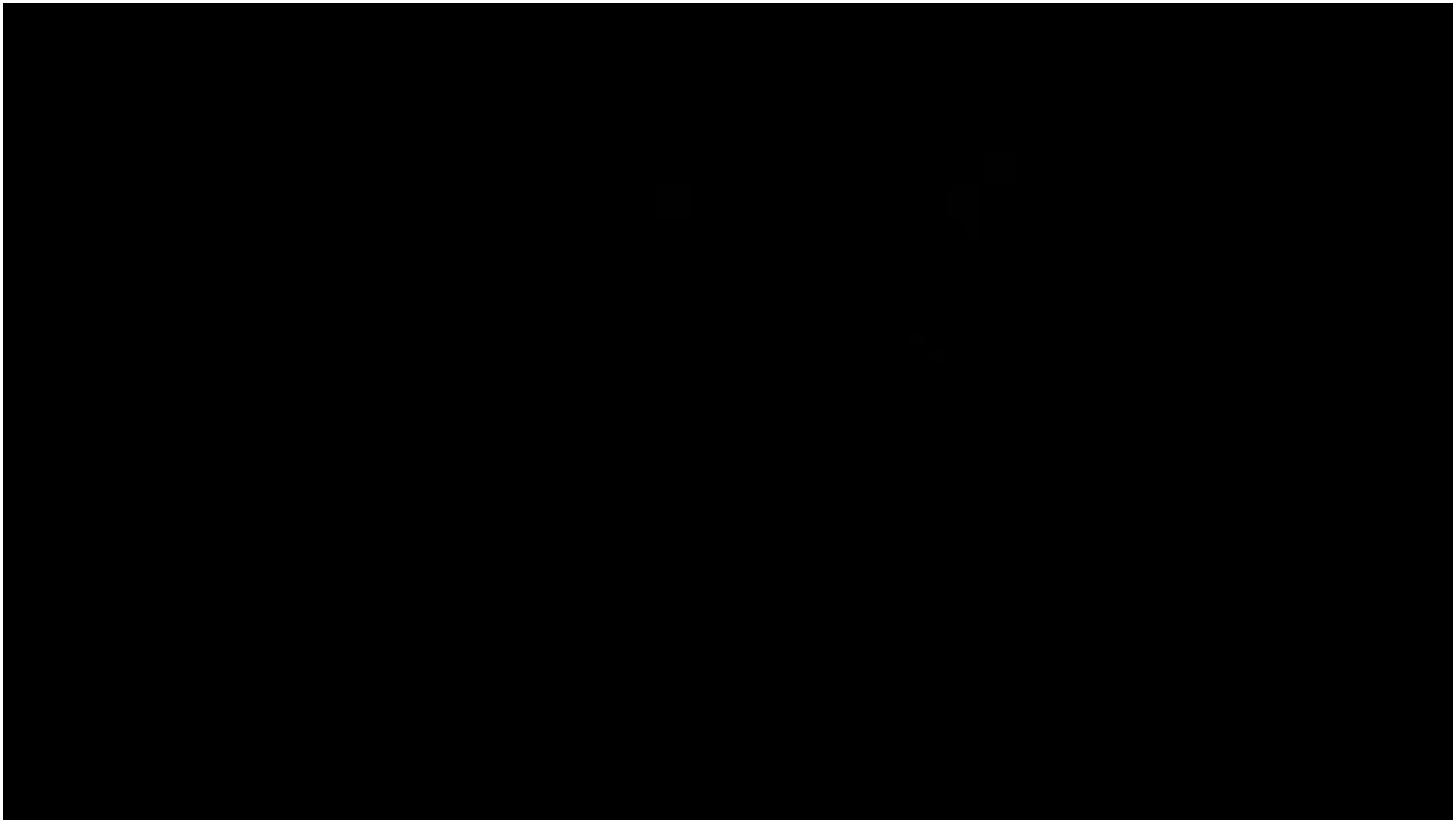
Biotechnology

7. Hoffman, J. S., Clark, P. U., Parnell, A. C. & He, F. Science 355, 276–279 (2017).

Base editor repairs gene of premature-ageing disease

Wilbert P. Vermeij & Jan H. J. Hoeijmakers

No cure exists for the lethal premature-ageing condition Hutchinson–Gilford progeria. A gene-editing tool – adenine base editors – offers a way to treat the condition in mice. Could this approach lead to an effective therapy? See p.608





https://www.rtve.es/play/videos/en-portada/increible-caso-sammy/16280740/